Thermal Emission from Geometrically Thin Accretion Disks: The Role of Black Hole Mass and Accretion Rate

[Author name(s)]

October 31, 2026

Abstract

We investigate how black hole mass and accretion rate shape the thermal continuum of a geometrically thin, optically thick accretion disk. The disk is modelled as a set of concentric rings, each radiating as a blackbody at the local effective temperature of the steady Newtonian thin-disk solution with a zero-torque inner boundary at the Schwarzschild innermost stable circular orbit, as implemented in the published `agnpy` package, with the Planck function and physical constants taken from `astropy`. The emission of all rings is summed numerically to obtain the specific luminosity L_ν and the flux F_ν received by an observer at a given distance and inclination. The numerical scheme reproduces the analytic bolometric luminosity to a fractional accuracy of 5×10^{-6} and converges as N^{-2} in the number of rings. At a fixed Eddington ratio $L/L_{\rm Edd} = 0.1$, a $10\,M_\odot$ black hole has a maximum disk temperature of $T_{\rm max} = 4.4 \times 10^6\,\mathrm{K}$ and a spectral peak at $\approx 0.94\,\mathrm{keV}$ (soft X-rays), while a $10^8\,M_\odot$ black hole has $T_{\rm max} = 7.9 \times 10^4\,\mathrm{K}$ and peaks at $\approx 740\,\text{Å}$ (far ultraviolet). The two peaks differ by exactly $(10^7)^{1/4} \approx 56$, reflecting the scaling $T_{\rm max} \propto (\dot{M}/M^2)^{1/4}$. The bolometric luminosity depends only on the accretion rate, $L = \dot{M}c^2/12$, so the luminosity and the spectral peak together determine both $\dot{M}$ and M. We discuss the limitations imposed by neglecting general-relativistic effects, non-blackbody radiative transfer and additional emission components.

Contents

1 Introduction **2**

2 Physical model **2**
2.1 Assumptions . . . 2
2.2 Radial temperature profile . . . 3
2.3 Luminosity and the Eddington limit . . . 3
2.4 Spectrum . . . 3

3 Numerical method **4**
3.1 Model implementation and software . . . 4
3.2 Ring discretisation . . . 4
3.3 Derived quantities . . . 5
3.4 Validation . . . 5

4 Results **6**
4.1 Temperature structure . . . 7
4.2 Stellar-mass versus supermassive black holes . . . 8
4.3 Dependence on black hole mass . . . 8
4.4 Dependence on accretion rate . . . 9
4.5 Scaling relations . . . 10
4.6 Observed flux: distance and inclination . . . 10

5 Discussion **11**
5.1 Connecting observables to accretion physics 11
5.2 Limitations 11

6 Conclusions **12**

A Code and reproducibility **13**

1 Introduction

Accretion onto black holes is among the most efficient ways of converting rest-mass energy into radiation. Black holes span more than eight orders of magnitude in mass, from stellar-mass objects of $\sim$ 5–20 $M_\odot$ in X-ray binaries to supermassive black holes of 10^6–$10^{10}\,M_\odot$ in active galactic nuclei (AGN). Despite this range, the same basic physics is thought to operate in both: gas with angular momentum settles into a disk, viscous stresses transport angular momentum outward and allow the gas to spiral inward, and the gravitational energy released is radiated from the disk surface.

The standard model for this process is the geometrically thin, optically thick disk of Shakura and Sunyaev [1], reviewed by Pringle [2] and Frank et al. [3]. Because the disk is optically thick, each annulus radiates approximately as a blackbody, and because the temperature falls with radius, the total spectrum is a superposition of blackbodies with a range of temperatures. This "multi-colour disk" spectrum is used to describe the thermal component of X-ray binaries in their soft state [4, 5] and the "big blue bump" seen in the optical–ultraviolet spectra of quasars [6, 7].

In this project we implement this model numerically and ask three questions:

1. How do the black hole mass M and accretion rate $\dot{M}$ set the disk temperature, the spectral peak and the total luminosity?
2. What flux would an observer at a given distance and inclination measure?
3. Why does the same physical model put the emission of stellar-mass black holes in the X-ray band and that of supermassive black holes in the ultraviolet?

Section 2 summarises the physical model, Section 3 describes the numerical method and its validation, Section 4 presents the results, and Section 5 connects them to observables and discusses the limitations of the model.

2 Physical model

2.1 Assumptions

We consider a steady, axisymmetric disk around a non-rotating black hole of mass M accreting at a constant rate $\dot{M}$. The disk is assumed to be

- *geometrically thin*, $H/r \ll 1$, so that it can be treated as a flat surface and radial pressure support is negligible;
- *optically thick*, so that each surface element radiates as a blackbody at its local effective temperature;
- *Keplerian and Newtonian*, with the inner edge at the innermost stable circular orbit (ISCO) of a Schwarzschild black hole, $r_{\rm in} = 6r_g$, where $r_g = GM/c^2$ is the gravitational radius.

Unless stated otherwise the outer radius is $r_{\rm out} = 10^5 r_g$.

2.2 Radial temperature profile

Conservation of mass and angular momentum in a steady thin disk, together with the assumption that the viscous torque vanishes at $r_{\rm in}$, gives the rate of energy dissipation per unit area of each face [1, 3]:

$$D(r) = \frac{3GM\dot{M}}{8\pi r^3}\left[1 - \left(\frac{r_{\rm in}}{r}\right)^{1/2}\right]. \tag{1}$$

In local thermal equilibrium this energy is radiated as a blackbody, so $\sigma T^4(r) = D(r)$ and

$$T(r) = T_*\left(\frac{r_{\rm in}}{r}\right)^{3/4}\left[1 - \left(\frac{r_{\rm in}}{r}\right)^{1/2}\right]^{1/4}, \qquad T_* = \left(\frac{3GM\dot{M}}{8\pi\sigma r_{\rm in}^3}\right)^{1/4}. \tag{2}$$

Far from the inner edge $T \propto r^{-3/4}$. The temperature vanishes at $r_{\rm in}$ and reaches its maximum at $r = \frac{49}{36} r_{\rm in}$, where

$$T_{\rm max} = \left[\frac{(36/49)^3}{7}\right]^{1/4} T_* \approx 0.488\, T_*. \tag{3}$$

Because $r_{\rm in} \propto M$, the temperature scale obeys

$$T_{\rm max} \propto \left(\frac{\dot{M}}{M^2}\right)^{1/4}. \tag{4}$$

2.3 Luminosity and the Eddington limit

Integrating $2D(r)$ over both faces from $r_{\rm in}$ to $r_{\rm out}$ gives the exact bolometric luminosity. With $X = r_{\rm out}/r_{\rm in}$,

$$L = \frac{3GM\dot{M}}{2r_{\rm in}}\left[\left(1 - \frac{1}{X}\right) - \frac{2}{3}\left(1 - X^{-3/2}\right)\right] \xrightarrow{X\to\infty} \frac{GM\dot{M}}{2r_{\rm in}} = \frac{\dot{M}c^2}{12}. \tag{5}$$

The radiative efficiency of this Newtonian model is therefore $\eta = 1/12$, and the luminosity depends on $\dot{M}$ alone. We parametrise the accretion rate through the Eddington ratio $\lambda = L/L_{\rm Edd}$, where

$$L_{\rm Edd} = \frac{4\pi GMm_p c}{\sigma_T} \approx 1.26 \times 10^{38}\,{\rm erg\,s^{-1}}\left(\frac{M}{M_\odot}\right) \tag{6}$$

is the luminosity at which radiation pressure on electrons balances gravity for ionised hydrogen. The accretion rate is then $\dot{M} = \lambda L_{\rm Edd}/(\eta c^2)$; for a $10\,M_\odot$ black hole at $\lambda = 1$ this is $\dot{M} = 1.68 \times 10^{19}\,{\rm g\,s^{-1}} = 2.7 \times 10^{-7}\,M_\odot\,{\rm yr^{-1}}$. At fixed λ, $\dot{M} \propto M$, so Eq. (4) gives $T_{\rm max} \propto M^{-1/4}$: *more massive black holes have cooler disks.*

2.4 Spectrum

Each annulus of area $dA = 2\pi r\, dr$ per face emits isotropically with specific intensity $B_\nu(T)$, the Planck function [8]

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT} - 1}. \tag{7}$$

The specific luminosity emitted from both faces is

$$L_\nu = 2\int_{r_{\rm in}}^{r_{\rm out}} \pi B_\nu(T(r))\, 2\pi r\, dr. \tag{8}$$

An observer at distance D who views the disk at inclination i (with $i = 0$ face-on) sees a projected area $dA\cos i$, so the received flux density is

$$F_\nu = \frac{\cos i}{D^2}\int_{r_{\rm in}}^{r_{\rm out}} B_\nu(T(r))\, 2\pi r\, dr = \frac{\cos i}{2\pi D^2}\, L_\nu. \tag{9}$$

The spectrum has three regimes. For $h\nu \ll kT(r_{\rm out})$ every annulus is in the Rayleigh–Jeans limit and $L_\nu \propto \nu^2$. For $h\nu \gg kT_{\rm max}$ the spectrum is cut off exponentially, as in the Wien tail. In between, the substitution $x = h\nu/kT(r)$ with $T \propto r^{-3/4}$ gives $r \propto x^{4/3}\nu^{-4/3}$, and Eq. (8) becomes

$$L_\nu \propto \nu^3 \int \frac{r\,dr}{e^x - 1} \propto \nu^3\,\nu^{-8/3} \int \frac{x^{5/3}\,dx}{e^x - 1} \propto \nu^{1/3}, \tag{10}$$

the characteristic signature of a multi-temperature disk.

3 Numerical method

3.1 Model implementation and software

Rather than re-deriving the disk model, we build on established, published software:

- **Disk model.** The temperature profile, Eq. (2), is taken from the `SSDisk` class of `agnpy` [9], an open-source, peer-reviewed Python package for the radiative processes of active galactic nuclei. `SSDisk` implements the Shakura and Sunyaev [1] disk with a zero-torque inner boundary, following Dermer and Menon [10]. agnpy describes a disk by $(M, L_{\rm disk}, \eta, r_{\rm in}, r_{\rm out})$; we pass $L_{\rm disk} = \eta\dot{M}c^2$ with $\eta = 1/12$, so agnpy's accretion rate equals ours exactly.
- **Planck function and constants.** $B_\nu(T)$ is evaluated with the `BlackBody` model of `astropy` [11], and all physical constants come from `astropy.constants` (CODATA 2018 and IAU 2015 values).

What this project adds is the integration over the disk (Section 3.2). agnpy can also compute the disk spectrum itself (`SSDisk.evaluate_multi_T_bb_sed`), using the same flux formula as Eq. (9), but it integrates over 100 *linearly* spaced radii. That is accurate for compact disks, the case agnpy was designed for, but for $r_{\rm out} = 10^5 r_g$ the grid spacing is $\sim 10^3 r_g$ and the hot inner disk is not resolved. We therefore use agnpy's temperature profile with our own logarithmically spaced ring sum, and verify the two against each other in Section 3.4.

The same Shakura–Sunyaev spectrum is also standard in X-ray astronomy: it is available in XSPEC [12] as `ezdiskbb` [13], which uses the same zero-torque profile, and in the simpler `diskbb` [4], which uses a pure $T \propto r^{-3/4}$ law.

3.2 Ring discretisation

The code (`accretion_disk/disk.py`) works in CGS units throughout. The disk is divided into N rings whose edges $r_0 = r_{\rm in} < r_1 < \cdots < r_N = r_{\rm out}$ are logarithmically spaced, so every decade in radius receives equal resolution. Ring k is assigned the temperature $T_k = T(\bar{r}_k)$ at its geometric-mean radius $\bar{r}_k = \sqrt{r_k r_{k+1}}$ and the one-face area $A_k = \pi(r_{k+1}^2 - r_k^2)$. Equations (8) and (9) become sums over rings,

$$L_\nu = 2\pi \sum_{k=0}^{N-1} B_\nu(T_k)\,A_k, \qquad F_\nu = \frac{\cos i}{D^2} \sum_{k=0}^{N-1} B_\nu(T_k)\,A_k, \tag{11}$$

which are evaluated for all frequencies at once as a matrix–vector product. The default is $N = 2000$. The Planck function is set to exactly zero for $h\nu/kT > 600$, where it is below 10^{-255} of its peak, to avoid floating-point overflow in the Wien tail.

Figure 1 illustrates the method: grouping the 2000 rings into eight zones shows that each zone contributes a nearly blackbody spectrum, while their sum is the broad multi-temperature disk spectrum.

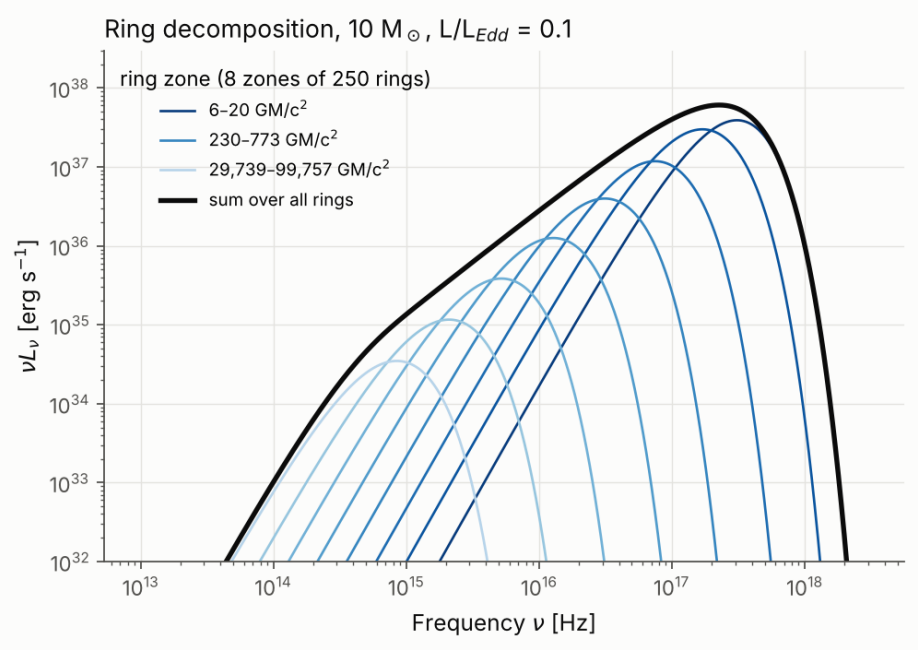


Figure 1: Decomposition of the disk spectrum of a $10\,M_\odot$ black hole at $\lambda = 0.1$. Coloured curves show the summed emission of eight zones of 250 rings each, from the hot inner zone (dark) to the cool outer zone (light); the black curve is the total. Each zone is close to a blackbody, but the total is much broader.

3.3 Derived quantities

The bolometric luminosity is computed directly from the rings as $L = 2\sigma \sum_k T_k^4 A_k$. The peak frequency ν_{peak} of νL_ν is found on a logarithmic grid of 4000 frequencies spanning 10^{-5} to 40 times kT_{max}/h; the grid maximum is refined by fitting a parabola to the three neighbouring points in $\log\nu$–$\log(\nu L_\nu)$ space. Scaling the grid with T_{max} gives the same relative resolution for every mass and accretion rate.

3.4 Validation

The implementation is checked by an automated test suite (`tests/test_disk.py`, 22 tests). Table 1 summarises the main checks against analytic results and against agnpy.

Table 1: Validation of the numerical model against analytic results and against agnpy.

Check	Result
$\pi \int B_\nu \, d\nu = \sigma T^4$	agrees to $< 10^{-4}$
Ring sum of σT^4 vs. Eq. (5)	5.4×10^{-6} ($N = 2000$)
$\int L_\nu \, d\nu$ vs. Eq. (5)	9.5×10^{-6}
Location and value of T_{max}	$r = \frac{49}{36} r_{\mathrm{in}}$, $T_{\mathrm{max}} = 0.488\,T_*$
Low-frequency slope	$L_\nu \propto \nu^{2.00}$
Intermediate slope ($r_{\mathrm{out}} = 10^8 r_g$)	$L_\nu \propto \nu^{0.33}$
Mass scaling at fixed λ	$\nu_{\mathrm{peak}} \propto M^{-0.2500}$
Accretion-rate scaling at fixed M	$\nu_{\mathrm{peak}} \propto \dot{M}^{0.2500}$, $L \propto \dot{M}^{1.000}$
Observer geometry	$F_\nu \propto \cos i / D^2$, face-on $F_\nu = L_\nu / 2\pi D^2$
$T(r)$ vs. closed form, Eq. (2)	identical to $< 10^{-10}$
νF_ν vs. agnpy SED ($r_{\mathrm{out}} = 50 r_g$)	agrees to $< 0.3\,\%$

Figure 2 shows the convergence of the ring sum. The error in the bolometric luminosity falls as N^{-2}, as expected for a midpoint-type rule; about 100 rings give 0.2 % accuracy, and the default

of 2000 rings gives errors well below 10^{-5}, negligible compared with the physical uncertainties of the model.

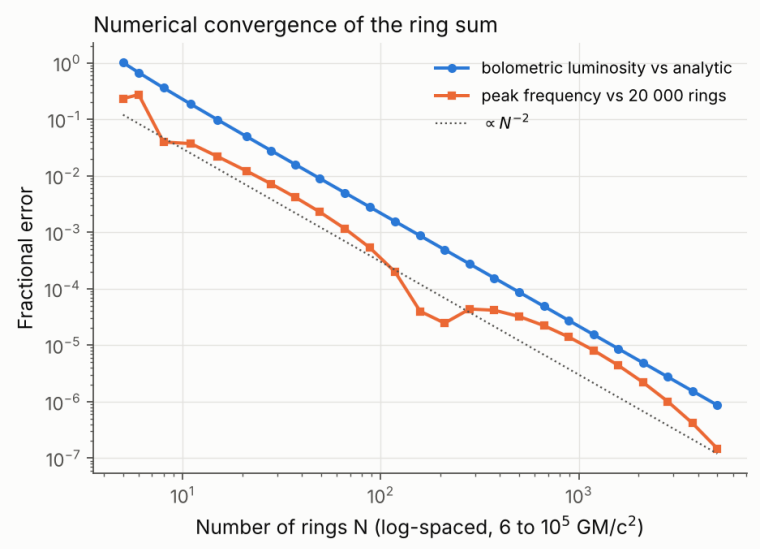


Figure 2: Fractional error of the bolometric luminosity (relative to Eq. 5) and of the peak frequency (relative to a 20 000-ring calculation) as a function of the number of rings N.

Figure 3 compares our ring sum with agnpy's own spectrum calculation. For compact disks ($r_{\rm out} \leq 50 r_g$), where agnpy's 100-point linear grid is adequate, the two agree to 0.1 % at all frequencies that carry significant flux. This confirms that our code computes the same Shakura–Sunyaev spectrum as the published package. As $r_{\rm out}$ grows, agnpy's grid becomes too coarse to resolve the inner disk: at $r_{\rm out} = 10^3 r_g$ it underestimates the flux at $h\nu = 10\,kT_{\rm max}$ by a factor of five, and at $r_{\rm out} = 10^5 r_g$ it misses the spectral peak entirely.

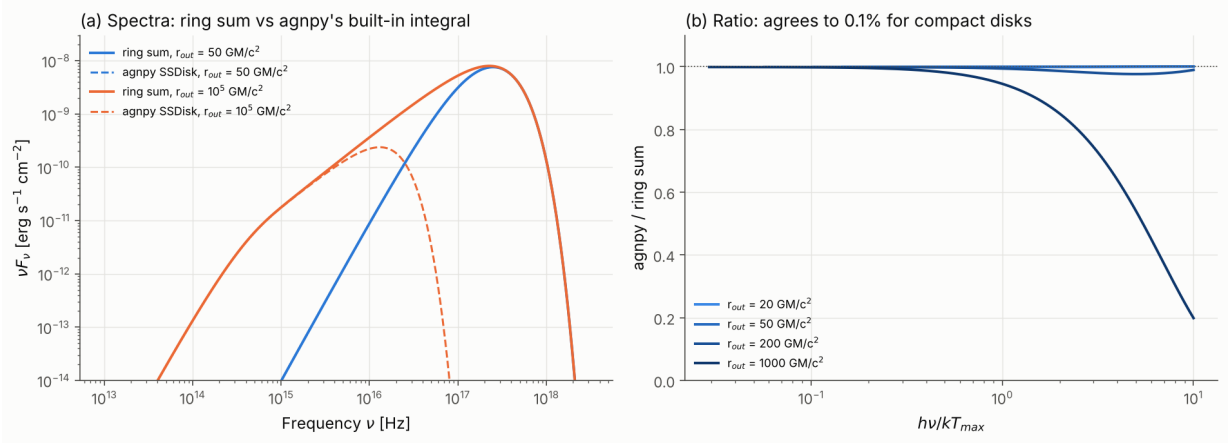


Figure 3: (a) νF_ν of a $10\,M_\odot$ disk at $\lambda = 0.1$ (8 kpc, $i = 60°$) from our ring sum (solid) and from agnpy's built-in integration (dashed), for $r_{\rm out} = 50 r_g$ (the curves overlap) and $r_{\rm out} = 10^5 r_g$. (b) Ratio of agnpy's result to the ring sum for several disk sizes.

4 Results

We adopt two representative systems, both accreting at $\lambda = 0.1$:

- a **stellar-mass black hole** of $10\,M_\odot$ at $D = 8$ kpc viewed at $i = 60°$, typical of a Galactic X-ray binary;

- a **supermassive black hole** of $10^8\,M_\odot$ at $D = 100\,\mathrm{Mpc}$ viewed at $i = 30°$, typical of a Seyfert nucleus or quasar.

Their properties at three Eddington ratios are listed in Table 2.

Table 2: Disk properties of the representative systems. $\nu_{\rm peak}$ and $E_{\rm peak} = h\nu_{\rm peak}$ refer to the peak of νL_ν; $F_{\rm bol}$ is the bolometric flux at the stated distance and inclination ($D = 8\,\mathrm{kpc}$, $i = 60°$ for $10\,M_\odot$; $D = 100\,\mathrm{Mpc}$, $i = 30°$ for $10^8\,M_\odot$).

M $[M_\odot]$	$L/L_{\rm Edd}$	$\dot{M}$ $[M_\odot\,\mathrm{yr}^{-1}]$	$L_{\rm bol}$ $[\mathrm{erg\,s}^{-1}]$	$T_{\rm max}$ [K]	$\nu_{\rm peak}$ [Hz]	$E_{\rm peak}$ [keV]	$F_{\rm bol}$ $[\mathrm{erg\,s}^{-1}\,\mathrm{cm}^{-2}]$
10	0.01	2.7×10^{-9}	1.26×10^{37}	2.49×10^{6}	1.28×10^{17}	0.53	1.6×10^{-9}
10	0.1	2.7×10^{-8}	1.26×10^{38}	4.42×10^{6}	2.27×10^{17}	0.94	1.6×10^{-8}
10	1	2.7×10^{-7}	1.26×10^{39}	7.86×10^{6}	4.04×10^{17}	1.67	1.6×10^{-7}
10^8	0.01	2.7×10^{-2}	1.26×10^{44}	4.42×10^{4}	2.27×10^{15}	0.0094	1.8×10^{-10}
10^8	0.1	2.7×10^{-1}	1.26×10^{45}	7.86×10^{4}	4.04×10^{15}	0.0167	1.8×10^{-9}
10^8	1	2.7	1.26×10^{46}	1.40×10^{5}	7.18×10^{15}	0.0297	1.8×10^{-8}

4.1 Temperature structure

Figure 4 shows the temperature profiles of the two systems. Expressed in units of r_g, the profiles have identical shapes: a steep rise from zero at the ISCO, a maximum at $8.2\,r_g$, and a $T \propto r^{-3/4}$ decline. Only the normalisation differs. The inner disk of the $10\,M_\odot$ black hole reaches $4.4\times10^6\,\mathrm{K}$, while that of the $10^8\,M_\odot$ black hole reaches only $7.9\times10^4\,\mathrm{K}$, even though it radiates 10^7 times more power. The ratio of the two temperatures is exactly $(10^7)^{1/4} = 56.2$, as required by Eq. (4). The supermassive disk can be cooler because its emitting area, $\propto M^2$, is 10^{14} times larger. The zero-torque boundary condition lowers the temperature at the maximum by a factor of $7^{1/4} \approx 1.6$ relative to the pure power law (dotted lines), and drives it to zero at $r_{\rm in}$.

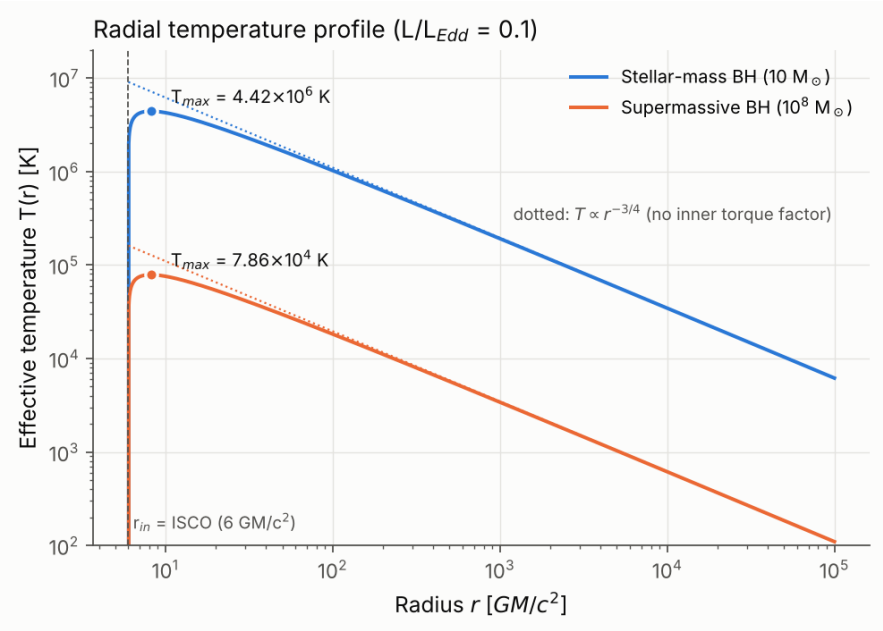


Figure 4: Effective temperature as a function of radius for a $10\,M_\odot$ and a $10^8\,M_\odot$ black hole at $\lambda = 0.1$. Dots mark $T_{\rm max}$ at $r = \frac{49}{36} r_{\rm in}$; dotted lines show the $T \propto r^{-3/4}$ power law without the inner-boundary factor.

4.2 Stellar-mass versus supermassive black holes

Figure 5 compares the spectra. The $10\,M_\odot$ disk peaks at $\nu_{\rm peak} = 2.27 \times 10^{17}$ Hz ($E \approx 0.94$ keV), in the soft X-ray band, which matches the thermal disk component observed in black hole X-ray binaries in their high/soft state [5]. The $10^8\,M_\odot$ disk peaks at $\nu_{\rm peak} = 4.04 \times 10^{15}$ Hz ($\lambda \approx 740$ Å), in the far ultraviolet, consistent with the big blue bump in quasar spectra [6, 7].

The spectral *shape* is the same in both cases when frequency is expressed in units of $kT_{\rm max}/h$: we find $h\nu_{\rm peak} = 2.466\,kT_{\rm max}$ for the peak of νL_ν and $h\nu = 0.80\,kT_{\rm max}$ for the peak of L_ν, independent of M and $\dot{M}$. The single model therefore produces characteristic frequencies that differ by a factor of 56 simply because $T_{\rm max} \propto M^{-1/4}$ at fixed Eddington ratio. Both spectra show the three regimes of Section 2: a Rayleigh–Jeans ν^2 tail below the temperature of the outer edge ($T(r_{\rm out}) \approx 6200$ K for the stellar disk and ≈ 110 K for the supermassive disk), a $\nu^{1/3}$ section ($\nu L_\nu \propto \nu^{4/3}$) at intermediate frequencies, and a Wien cutoff above $\nu_{\rm peak}$.

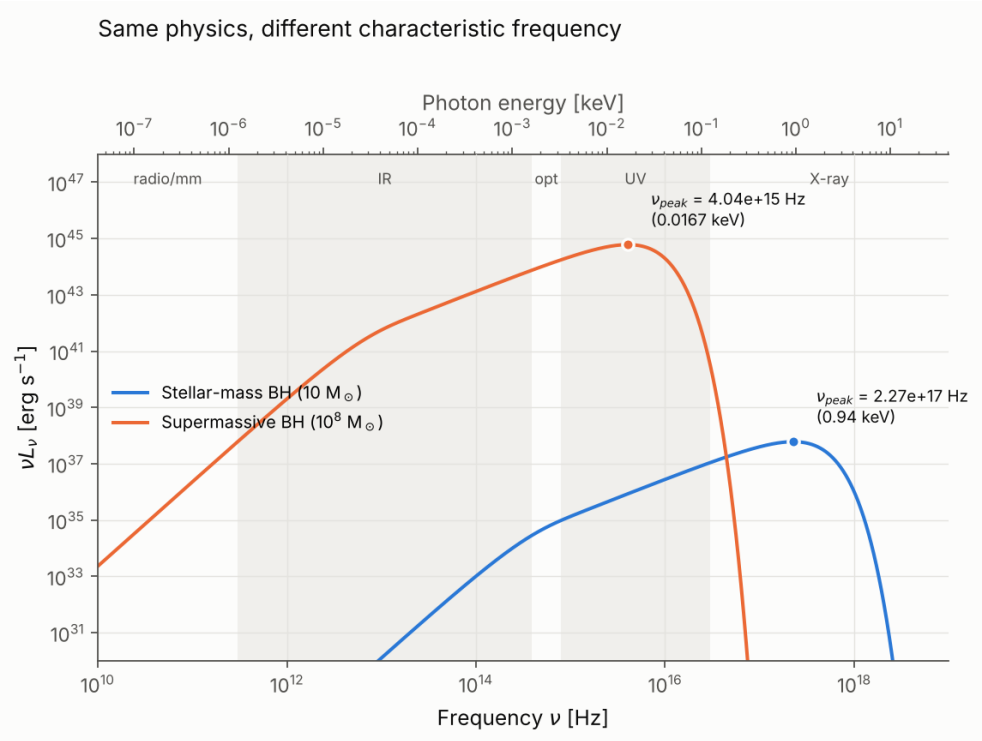


Figure 5: Spectral energy distributions νL_ν of the $10\,M_\odot$ and $10^8\,M_\odot$ disks at $\lambda = 0.1$. Shaded bands mark the IR and UV regions; the top axis gives the photon energy.

4.3 Dependence on black hole mass

Figure 6 shows a sequence of masses from 10 to $10^9\,M_\odot$ at fixed $\lambda = 0.1$. Increasing the mass raises the luminosity in proportion ($L \propto \dot{M} \propto M$) and moves the peak to lower frequency, so the peaks lie along the locus $\nu L_\nu \propto \nu_{\rm peak}^{-4}$ (dashed line). A power-law fit to the peak frequencies gives a slope of -0.2500, in exact agreement with $\nu_{\rm peak} \propto M^{-1/4}$. Over this mass range the peak moves from the X-ray band, through the extreme ultraviolet, into the far ultraviolet.

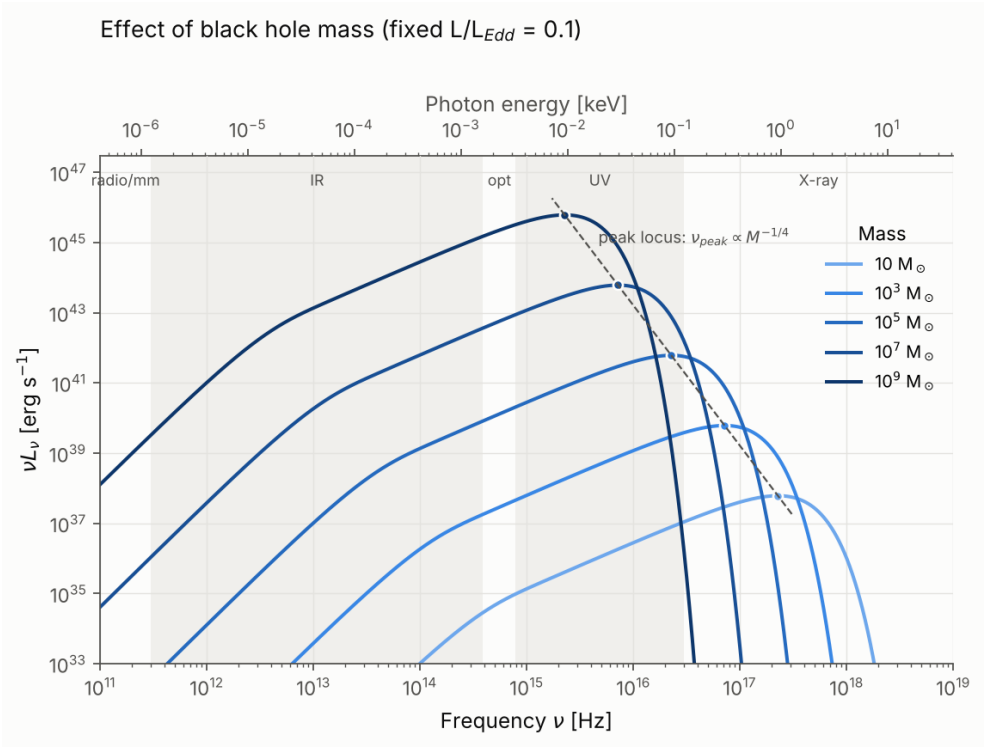


Figure 6: Disk spectra for black hole masses of 10, 10^3, 10^5, 10^7 and $10^9\,M_\odot$ at $\lambda = 0.1$. Dots mark the peak of each spectrum; the dashed line traces the peak locus.

4.4 Dependence on accretion rate

Figure 7 shows the effect of varying the accretion rate from $\lambda = 0.01$ to 1 at fixed mass. The luminosity increases in proportion to $\dot{M}$, but the peak moves only as $\dot{M}^{1/4}$: a hundredfold increase in accretion rate shifts $\nu_{\rm peak}$ by a factor of only $100^{1/4} \approx 3.2$ (from 0.53 to 1.67 keV for the stellar-mass disk). In the $\nu^{1/3}$ regime the spectra at different $\dot{M}$ are parallel, with $L_\nu \propto \dot{M}^{2/3}$ at fixed frequency.

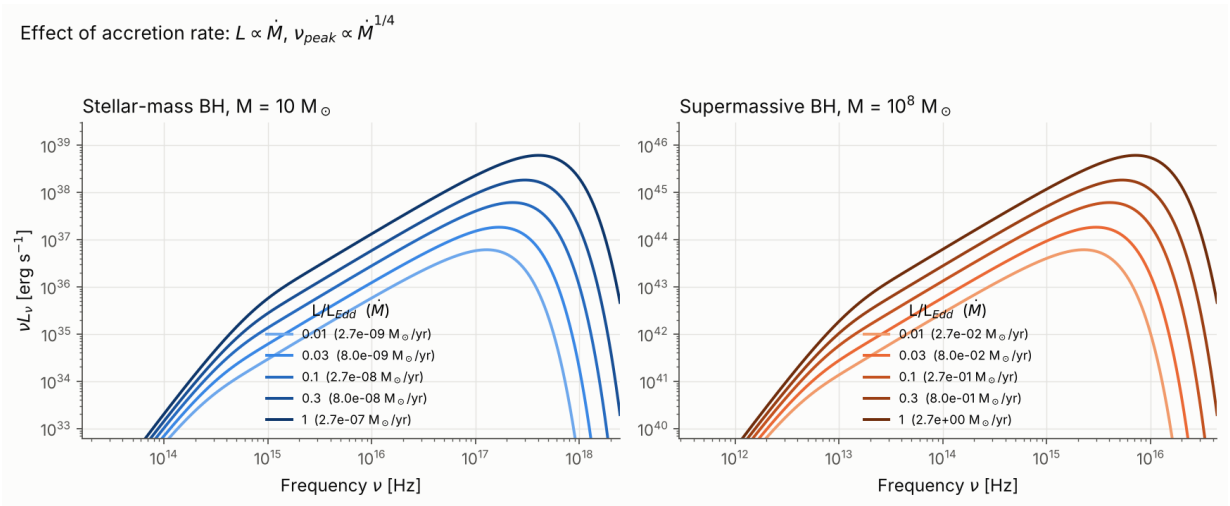


Figure 7: Disk spectra for Eddington ratios $\lambda = 0.01$–1 for the $10\,M_\odot$ (left) and $10^8\,M_\odot$ (right) black holes. The legend gives the corresponding accretion rate.

4.5 Scaling relations

Figure 8 summarises the scaling laws. At fixed Eddington ratio $\nu_{\rm peak} \propto M^{-1/4}$ (panel a); at fixed mass $\nu_{\rm peak} \propto \dot{M}^{1/4}$ (panel b); and the bolometric luminosity lies on the single relation $L = \dot{M}c^2/12$ for every mass (panel c). Combining these, the model results can be written compactly as

$$T_{\rm max} \approx 4.4 \times 10^6\,{\rm K} \left(\frac{M}{10\,M_\odot}\right)^{-1/4} \left(\frac{\lambda}{0.1}\right)^{1/4}, \tag{12}$$

$$\nu_{\rm peak} \approx 2.3 \times 10^{17}\,{\rm Hz} \left(\frac{M}{10\,M_\odot}\right)^{-1/4} \left(\frac{\lambda}{0.1}\right)^{1/4}. \tag{13}$$

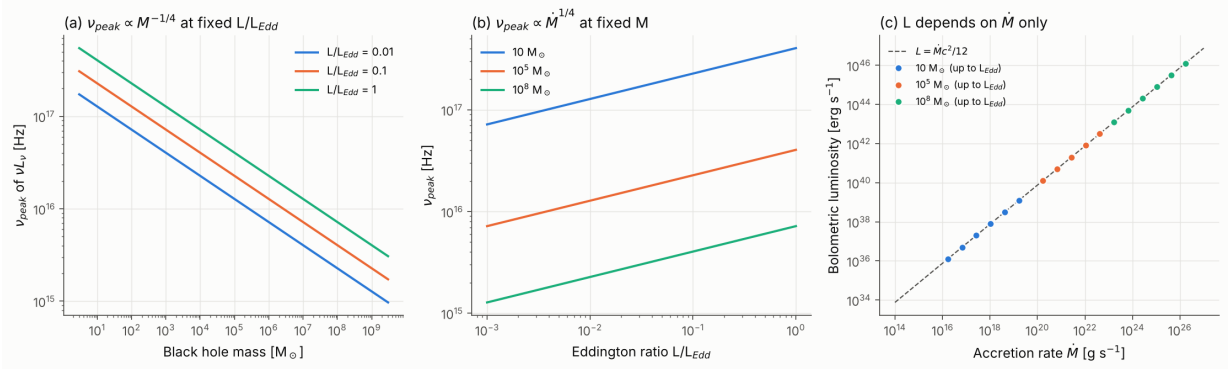


Figure 8: (a) Peak frequency versus black hole mass at three Eddington ratios. (b) Peak frequency versus Eddington ratio at three masses. (c) Bolometric luminosity versus accretion rate; points for all masses lie on $L = \dot{M}c^2/12$.

4.6 Observed flux: distance and inclination

Figure 9 shows the flux received by observers of the two fiducial systems at several inclinations. At $\lambda = 0.1$ the X-ray binary at 8 kpc and $i = 60°$ has a bolometric flux of $1.6 \times 10^{-8}\,{\rm erg\,s^{-1}\,cm^{-2}}$, typical of a bright Galactic X-ray source. The AGN at 100 Mpc and $i = 30°$ has $1.8 \times 10^{-9}\,{\rm erg\,s^{-1}\,cm^{-2}}$, its far greater luminosity being largely offset by its 1.25×10^4 times greater distance.

In this non-relativistic model inclination affects only the normalisation through $\cos i$; the spectral shape is unchanged. Going from face-on to $i = 85°$ reduces the flux by a factor of 11. An observer who assumes isotropic emission and infers $L_{\rm iso} = 4\pi D^2 F$ obtains $L_{\rm iso} = 2\cos i\, L$, which overestimates the true luminosity by up to a factor of two for face-on disks ($L_{\rm iso}/L = 1.73$ for the AGN at $i = 30°$) and underestimates it for $i > 60°$.

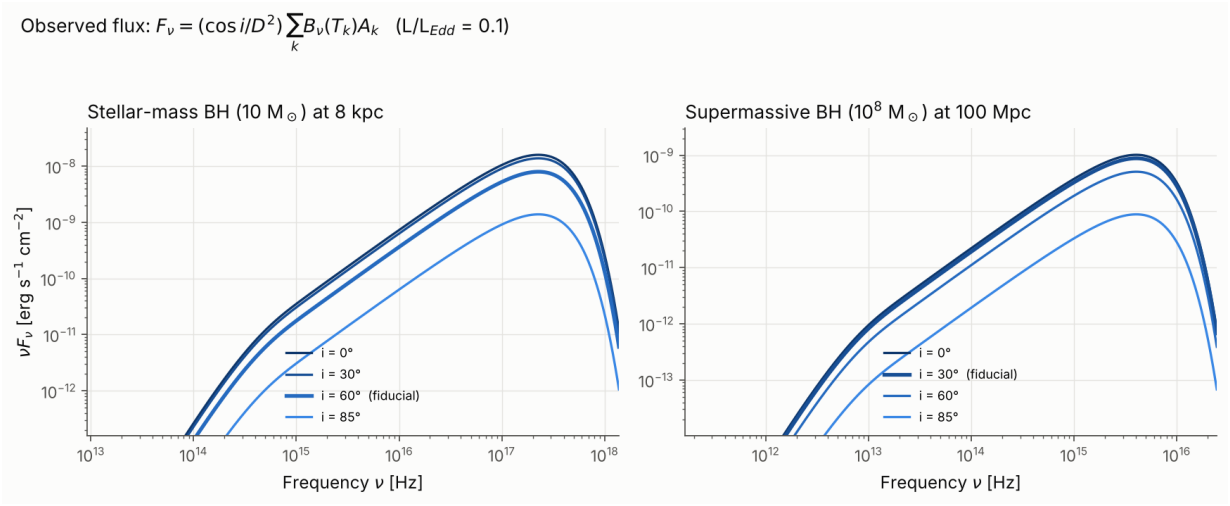


Figure 9: Observed νF_ν for the $10\,M_\odot$ black hole at 8 kpc (left) and the $10^8\,M_\odot$ black hole at 100 Mpc (right), both at $\lambda = 0.1$, for inclinations of 0°, 30°, 60° and 85°. The fiducial inclination is drawn with a thicker line.

5 Discussion

5.1 Connecting observables to accretion physics

The results show how the two basic parameters of the accretion flow map onto observable spectral properties.

- **The bolometric luminosity measures the accretion rate.** Because $L = \eta \dot{M} c^2$ with a mass-independent efficiency, a luminosity measurement (from the flux, distance and inclination) gives $\dot{M}$ directly, but carries no information about M.
- **The spectral peak measures the temperature scale.** The peak frequency gives $T_{\max} \propto (\dot{M}/M^2)^{1/4}$, a combination of mass and accretion rate.
- **Together they determine the mass.** Eliminating $\dot{M}$ between $L = \dot{M}c^2/12$ and Eq. (2) gives

$$M = \left[\frac{3G\dot{M}}{8\pi\sigma T_*^4\,(6G/c^2)^3}\right]^{1/2} \propto L^{1/2}\,T_{\max}^{-2}. \tag{14}$$

  Applying this inversion to synthetic spectra recovers the input mass to 0.03 % for both the stellar-mass and supermassive cases (see the accompanying notebook). This is the principle behind continuum-fitting estimates of black hole mass and, with relativistic models, spin.

The weak, fourth-root dependence of temperature on both parameters explains why disks around black holes of similar type have similar characteristic energies: stellar-mass disks cluster around 0.5 keV to 2 keV, and AGN disks peak in the ultraviolet. It also means that a large change in accretion rate produces only a modest shift in the spectral peak but a large change in luminosity.

5.2 Limitations

The model captures the essential physics of thermal disk emission, but several effects that matter in comparisons with real data are neglected.

General-relativistic disk structure. The Newtonian temperature profile should be replaced by the relativistic solution of Novikov and Thorne [14]. More importantly, black hole spin moves the ISCO from $6r_g$ (non-rotating) to as close as $1r_g$ for a maximally rotating black hole with a prograde disk [15]. This increases the radiative efficiency from $\approx 5.7\,\%$ (Schwarzschild, compared with our Newtonian $1/12 \approx 8.3\,\%$) to as much as $\approx 42\,\%$, makes the inner disk hotter, and shifts ν_{peak} upward. Our results therefore correspond most closely to slowly rotating black holes.

Relativistic photon propagation. Near the ISCO the orbital speed is $\sim 0.4c$, so Doppler boosting, gravitational redshift and light bending are significant. They make the observed spectral shape, not only its normalisation, depend on inclination: edge-on disks appear harder and brighter in their high-energy tail than our $\cos i$ scaling predicts.

Departures from blackbody emission. In the hot inner regions of stellar-mass disks electron scattering dominates over absorption, so the emergent spectrum is a diluted, hardened blackbody, usually described by a colour-correction factor $f_{\mathrm{col}} \approx 1.7$ [16]. Atomic absorption edges and limb darkening also modify the local spectra, particularly in AGN disks.

Additional emission components. Real spectra contain components beyond the thermal disk: a hot, Comptonising corona that produces power-law X-ray emission; reflection features such as the Fe Kα line; synchrotron emission from jets; reprocessing of X-rays in the outer disk (irradiation); and emission from the companion star, the broad-line region, the dusty torus and the host galaxy.

Range of validity of the thin-disk solution. The assumption $H/r \ll 1$ breaks down as L approaches L_{Edd}, where radiation pressure puffs up the inner disk and advection becomes important (slim disks), and at very low accretion rates, where the flow may become radiatively inefficient. AGN disks also become self-gravitating beyond $\sim 10^3$–$10^4 r_g$, so the adopted $r_{\mathrm{out}} = 10^5 r_g$ affects their predicted infrared emission.

Propagation effects. We neglect interstellar and intrinsic absorption, which removes most extreme-ultraviolet photons and hides the peak of AGN disks from direct observation, as well as cosmological redshift and time variability.

6 Conclusions

We have modelled the thermal emission of a geometrically thin, optically thick accretion disk using the standard Shakura–Sunyaev model as implemented in the published `agnpy` package, with `astropy` for the Planck function and physical constants, and by summing the blackbody emission of concentric rings. The ring sum agrees with agnpy's own spectrum to 0.1 % where the latter is valid, converges as N^{-2} and reproduces the analytic luminosity to better than 10^{-5}. Our main findings are:

1. The disk temperature obeys $T_{\mathrm{max}} \propto (\dot{M}/M^2)^{1/4}$, so at a fixed Eddington ratio more massive black holes have cooler disks, $T_{\mathrm{max}} \propto M^{-1/4}$.

2. The same model puts the thermal peak of a $10\,M_\odot$ black hole at $\sim 1\,\mathrm{keV}$ (soft X-rays) and that of a $10^8\,M_\odot$ black hole at $\sim 740\,\text{Å}$ (far UV), a factor of 56 apart in frequency.

3. The bolometric luminosity, $L = \dot{M}c^2/12$, depends only on the accretion rate; the spectral peak, $h\nu_{\mathrm{peak}} \approx 2.47\,kT_{\mathrm{max}}$, moves only as $\dot{M}^{1/4}$.

4. The luminosity and the peak frequency together determine both $\dot{M}$ and M.

5. For an observer, the flux scales as $\cos i / D^2$, and assuming isotropic emission biases the inferred luminosity by a factor of $2\cos i$.

The most important extensions of this work would be to include the relativistic Novikov–Thorne temperature profile with black hole spin, relativistic ray tracing, a colour-correction factor, and a Comptonised coronal component.

A Code and reproducibility

All results in this report can be reproduced from the project repository:

`accretion_disk/disk.py`	`ThinDisk` (agnpy `SSDisk` + ring sum)
`scripts/run_analysis.py`	generates all figures and `results/summary.csv`
`tests/test_disk.py`	validation tests (`python -m pytest`)
`notebooks/accretion_disk_walkthrough.ipynb`	step-by-step walkthrough

The code requires Python 3 with `numpy`, `scipy`, `matplotlib`, `astropy` and `agnpy` (see `requirements.txt`). A minimal example:

```
from accretion_disk import ThinDisk, constants as C
disk = ThinDisk.from_eddington(mass_msun=10, eddington_ratio=0.1)
nu   = disk.frequency_grid()
L_nu = disk.luminosity_nu(nu)                                  # erg/s/Hz
F_nu = disk.flux_nu(nu, distance=8*C.kpc, inclination_deg=60)  # erg/s/cm^2/Hz
print(disk.T_max, disk.peak_frequency(), disk.bolometric_luminosity())
print(disk.ssdisk)      # the underlying agnpy.targets.SSDisk object
```

References

[1] N. I. Shakura and R. A. Sunyaev. Black holes in binary systems. observational appearance. *Astronomy and Astrophysics*, 24:337–355, 1973.

[2] J. E. Pringle. Accretion discs in astrophysics. *Annual Review of Astronomy and Astrophysics*, 19:137–162, 1981.

[3] J. Frank, A. King, and D. Raine. *Accretion Power in Astrophysics*. Cambridge University Press, Cambridge, 3rd edition, 2002.

[4] K. Mitsuda, H. Inoue, K. Koyama, et al. Energy spectra of low-mass binary x-ray sources observed from tenma. *Publications of the Astronomical Society of Japan*, 36:741–759, 1984.

[5] R. A. Remillard and J. E. McClintock. X-ray properties of black-hole binaries. *Annual Review of Astronomy and Astrophysics*, 44:49–92, 2006.

[6] G. A. Shields. Thermal continuum from accretion disks in quasars. *Nature*, 272:706–708, 1978.

[7] M. A. Malkan. The ultraviolet excess of luminous quasars. ii. evidence for massive accretion disks. *The Astrophysical Journal*, 268:582–590, 1983.

[8] G. B. Rybicki and A. P. Lightman. *Radiative Processes in Astrophysics*. Wiley, New York, 1979.

[9] C. Nigro, J. Sitarek, P. Gliwny, D. Sanchez, A. Tramacere, and M. Craig. agnpy: An open-source python package modelling the radiative processes of jetted active galactic nuclei. *Astronomy and Astrophysics*, 660:A18, 2022.

[10] C. D. Dermer and G. Menon. *High Energy Radiation from Black Holes: Gamma Rays, Cosmic Rays, and Neutrinos*. Princeton University Press, Princeton, 2009.

[11] Astropy Collaboration. The astropy project: sustaining and growing a community-oriented open-source project and the latest major release (v5.0) of the core package. *The Astrophysical Journal*, 935:167, 2022.

[12] K. A. Arnaud. Xspec: The first ten years. In *Astronomical Data Analysis Software and Systems V*, volume 101 of *ASP Conference Series*, page 17, 1996.

[13] E. R. Zimmerman, R. Narayan, J. E. McClintock, and J. M. Miller. Multitemperature blackbody spectra of thin accretion disks with and without a zero-torque inner boundary condition. *The Astrophysical Journal*, 618:832–844, 2005.

[14] I. D. Novikov and K. S. Thorne. Astrophysics of black holes. In C. DeWitt and B. S. DeWitt, editors, *Black Holes (Les Astres Occlus)*, pages 343–450. Gordon and Breach, New York, 1973.

[15] J. M. Bardeen, W. H. Press, and S. A. Teukolsky. Rotating black holes: locally nonrotating frames, energy extraction, and scalar synchrotron radiation. *The Astrophysical Journal*, 178: 347–370, 1972.

[16] T. Shimura and F. Takahara. On the spectral hardening factor of the x-ray emission from accretion disks in black hole candidates. *The Astrophysical Journal*, 445:780–788, 1995.